

Rean Clive Fernandes

Freiburg im Breisgau, DE
Machine Learning Engineer

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PROFILE

I am a passionate ML researcher/engineer with experience spanning computer vision, robotics, and LLMs. I specialize in solving complex, open-ended problems and translating them into tangible applications. My commitment to reproducible research and clean code drives me to leverage my skills for impactful, real-world challenges.

PROFESSIONAL EXPERIENCE

- **Research Associate: Agentic LLMs for Data Science** Freiburg, DE
Machine Learning Lab, Albert-Ludwigs-Universität Oct 2024 - Present
 - **Agentic LLMs:** Leading development of Agentic LLM pipelines for autonomous augmentation of tabular data.
 - **Auto Feature Engineering:** Prototyping tabular feature engineering methods with LLM integration.
 - **Reinforcement Learning:** Researching multi armed bandits as decision algorithms for data augmentation.
 - **Knowledge Graphs:** Designing and analyzing novel prompting methods to query knowledge graphs for LLMs.
- **Research Assistant: Data Curation for Language Model Training** Freiburg, DE
Machine Learning Lab, Albert-Ludwigs-Universität Apr 2024 - Oct 2024
 - **Data Distillation:** Engineered automated Python-based data processing workflows for LLM training, enhancing Llama 2 7B task performance by 20%.
 - **LLM Benchmarking:** Developed comprehensive evaluation pipelines for Llama and Mistral models on diverse unstructured legal text data
 - **Data Curation:** Curated and cleaned specialized datasets (500+ examples) to improve LLM training quality.
- **Teaching Assistant: Foundations of Deep Learning** Freiburg, DE
Albert-Ludwigs-Universität Oct 2023 - Sep 2024
 - Mentored students in understanding complex Deep Learning concepts and practical implementations using Python and PyTorch.
 - Developed supplementary materials and facilitated collaborative learning sessions.

EDUCATION

- **M.Sc. Embedded Engineering (Specialization: Artificial Intelligence)** Apr 2021 - Sep 2024
Albert-Ludwigs-Universität, Freiburg GPA: 2.3
 - **Thesis:** Supervised Fine-tuning of Open LLMs for Law: Training and Evaluation for Performance in Legal Examinations (Grade: 1.0)
 - Focus: Deep Learning, Natural Language Processing, Control Theory, Robotics, Optimization, Computer Vision.
- **B.Engg. Electronics and Communication Engineering** Jul 2016 - Jul 2020
NMAM Institute of Technology, Karnataka, India GPA: 8.5/10
 - Focus: Advanced Engineering Mathematics, Signal Processing, Embedded System Design, Control Theory.

SELECTED PROJECTS

- **Master Thesis: Advanced LLM Fine-Tuning & Pipeline Automation for Legal Reasoning** Mar 2024 - Sep 2024
Led end-to-end development of LLM solutions, achieving 3-4x performance gains by fine-tuning Llama 2 7B (Q-LoRa) on US Bar Exam questions. Engineered automated data distillation (Llama 3 70B, IRAC method) and evaluation pipelines (Python, PyTorch, Hugging Face Transformers on HPC). Created interactive visualizations (Streamlit, Plotly). Grade: 1.0
 - Key Skills: LLM Fine-tuning, Data curation and distillation, Pipeline Automation, Prompt Engineering, Python, PyTorch, Hugging Face, HPC
- **Smart SLURM Job Handler** Personal Project
Developed a Python-based automated job management tool with SQLite for HPC SLURM clusters, demonstrating practical MLOps skills in pipeline automation and resource optimization.
 - Key Skills: Python, SQLite, SLURM, HPC, Automation, MLOps.
- **Interactive LLM Q&A Bias Analysis with Streamlit and Plotly** Personal Project
An interactive Streamlit dashboard for analyzing potential answer selection biases in multiple-choice questions by LLM models. It compares predicted vs. ground truth distributions and visualizes label confusion. Used in my paper for the Multi-state Bar Exam to analyze supervised fine-tuning effects.
 - Key Skills: Streamlit, Plotly, ML Bias analysis

•Non-Linear Model Predictive Control for Crazyflie 2.1

Academic Project (M.Sc.)

Implemented and comparatively analyzed direct collocation and direct multiple shooting NMPC methods for a nano quadcopter. Involved system modeling, simulation, and performance analysis.

– Key Skills: Python, CasADi, Non-Linear MPC, Control System Design, ODE simulation

•Optimal Control of a Simulated Furuta Pendulum

Academic Project (M.Sc.)

Designed and simulated MPC controllers (Direct Multiple Shooting, Direct Collocation, Real-Time Iteration) using CasADi on MATLAB for swing-up and stabilization of a Furuta pendulum.

– Key Skills: MATLAB, CasADi, Model Predictive Control (MPC), Numerical Simulation, Control System Design.

•Analysis of Trust Region Optimization Methods

Academic Project (M.Sc.)

Implemented and comparatively analyzed various Trust Region sub-problem solution methods (Cauchy Point, Dog-leg, Conjugate Gradient) for non-linear optimization using Python and CasADi.

– Key Skills: Python, CasADi, Numerical Optimization

•Skill-based Model-based Reinforcement Learning (Research Contribution)

Academic Research Group (M.Sc.)

Contributed to research on combining model-based RL with skill-based approaches for improved sample efficiency and long-horizon task performance (conceptual work and discussions presented in poster format).

– Key Concepts: Reinforcement Learning (RL), Model-Based RL, Variational Autoencoders (VAEs)

PUBLICATIONS

•Rean Fernandes, Andre Biedenkapp, Frank Hutter, & Noor Awad

Preprint

"A Llama walks into the 'Bar': Efficient Supervised Fine-Tuning for Legal Reasoning in the Multi-state Bar Exam." 

TECHNICAL SKILLS

AI & LLMs: LLM Development (Fine-tuning: Q-LoRa, LoRA, SFT; Model Customization, Evaluation), LLM Pipelines (End-to-End, Automation), Dataset Distillation & Curation, Prompt Engineering.

ML Frameworks: PyTorch, Hugging Face (Transformers, Datasets, Tokenizers), LangChain, LangGraph, Scikit-learn.

Programming & Data: Python (Expert; Data/ML pipelines), SQL, Pandas, NumPy, MATLAB (Advanced), CasADi.

MLOps & Cloud: MLOps Principles (Git, Weights & Biases, SLURM), CI/CD (foundational), AWS.

Software & Tools: Docker, Git/GitHub, Linux/Unix, Jupyter Notebooks, VS Code, Streamlit, Plotly.

Languages: English (Fluent), German (A2, actively learning)

REFERENCES

•**Prof. Dr. Frank Hutter**, Professor, Machine Learning Lab, Albert-Ludwigs-Universität, fh@cs.uni-freiburg.de

•**Dr. André Biedenkapp** Post-Doctoral Research Fellow, Machine Learning Lab, Albert-Ludwigs-Universität, biedenka@cs.uni-freiburg.de

•**Dr. Noor Awad** Post-Doctoral Research Fellow, Machine Learning Lab, Albert-Ludwigs-Universität, awad@cs.uni-freiburg.de